DESIGN THINKING METHODOLOGY: CUSTOMER JOURNEY MAP

India's Agricultural Crop Production Analysis using Tableau

1. PROJECT & INSTITUTIONAL FRAMEWORK

This document outlines the end-to-end user experience and interaction architecture for the crop production analytical platform developed as part of the technical curriculum requirement.

Project Title:	India's Agricultural Crop Production Analysis using Tableau
Domain & Tool:	Data Analytics | Tableau Public
Internship Platform:	SmartBridge – SkillWallet (Final Year Capstone Submission)
Dataset Scope:	Indian Agricultural Crop Production Dataset (1997–2021)
Primary Persona:	Agricultural Officer / Government Policy Maker / Data Analyst

2. COMPREHENSIVE CUSTOMER JOURNEY MAP MATRIX

Dimension	1. Awareness	2. Need ID	3. Exploration	4. Analysis	5. Decision Making	6. Outcome
User Goal	Find an efficient method to interpret national crop data.	Identify specific parameters (yield, season, state area) to compare performance.	Navigate dashboard seamlessly using filters, maps, and dynamic charts.	Isolate state-wise variations, seasonal yields, and historical production trends.	Formulate data-backed agricultural policies and optimize resource allocation.	Establish an automated, accurate, and rapid reporting and strategy system.
User Actions	• Reviews complex historical spreadsheets. • Searches for visualization frameworks.	• Formulates requirements for cross-parameter tracking. • Selects Tableau Public.	• Launches Tableau Dashboard. • Interacts with geographic maps & KPI cards.	• Filters data by state/season. • Runs multi-variable yield trend analysis.	• Downloads tailored visual insights. • Integrates charts into policy papers.	• Replaces old legacy sheets. • Deploys automated insight model successfully.
User Thoughts	"Sifting through thousands of manual spreadsheet rows is highly error-prone."	"I need a single screen to correlate seasonal changes with regional crop yields."	"The interface is intuitive; maps instantly show high vs low yield regions."	"Certain states show massive drops in specific seasons. Let me check the exact cause."	"This clear visualization leaves zero ambiguity for the review committee."	"This saves dozens of working hours every week and improves accuracy tenfold."

Dimension	1. Awareness	2. Need ID	3. Exploration	4. Analysis	5. Decision Making	6. Outcome
User Emotions	😟 **Overwhelmed & Frustrated**	🔍 **Determined & Hopeful**	💡 **Curious & Engaged**	🎓 **Analytical & Absorbed**	💪 **Confident & Empowered**	🤩 **Satisfied & Accomplished**
Pain Points	• High data latency. • Hidden patterns remain completely unseen.	Defining consistent comparison metrics without analytical software.	Initial familiarity curve with multiple active filters.	Complex regional anomalies require deep filtering layers.	Formatting or exporting static charts for formal reports.	Managing data source refreshes as newer annual stats emerge.
Opportunities / Improvements	Adopt centralized Tableau dashboards instead of static spreadsheets for easier agricultural data analysis.	Create uniform data definitions for area, production, and yield.	Add quick-start inline tooltips or guided tutorials for first-time users.	Implement automated anomaly detection alert callouts in Tableau.	Include one-click automated PDF/ PPT summary report generators.	Update the Tableau dashboard whenever new yearly agricultural data becomes available to keep the analysis current.

3. IMPLEMENTATION AND WORKFLOW INSIGHTS

By understanding each stage of the user's journey, the Tableau Public solution simplifies agricultural data analysis and helps users explore production trends more efficiently. Transitioning from spreadsheet-based analysis to interactive maps, seasonal trend charts, KPI cards, and dashboards enables agricultural stakeholders to analyse agricultural production data from 1997–2021 quickly and efficiently.